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Paleomagnetic results from the lower Mesozoic Daedong Supergroup in the Gyeonggi Block, Korean Peninsula: an eastern extension of the South China Block

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Abstract

Lower Mesozoic lake deposits in the Daedong Supergroup of the Gyeonggi Block were collected at 38 sites for paleomagnetic study. A remanent magnetization component with unblocking temperatures of 635–660°C (component A) was isolated from four sites. The directions of this component with both normal and reversed polarities yielded positive fold and reversal tests at the 95% confidence level. Component A is interpreted to be of primary origin of Late Triassic time. The mean direction of component A ($D = 82.2°$, $I = 23.1°$ with $\alpha_{95} = 4.1°$), together with previously reported primary directions, yields a pole representative for the Gyeonggi Block at 31.9°N, 220.2°E ($A_{95} = 12.7°$). The paleomagnetic pole shows good agreement with that for the South China Block, suggesting that the Gyeonggi Block has been part of the South China Block. The Gyeonggi Block, as the eastern part of the South China Block, is regarded to have thrust to its position at the present Korean Peninsula during the collision between the North and South China Blocks. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Korea; paleomagnetism; tectonics

1. Introduction

The Korean Peninsula is situated at the eastern ends of the major tectonic blocks comprising East Asia (North and South China Blocks, see Fig. 1a), and is composed of three tectonic units with Precambrian basement, namely the Nangrim, Gyeonggi and Ryeongnam Blocks from north to south [1,2] (Fig. 1b). The former two blocks are bounded by the Imjingang Belt, and the latter two by the Ogcheon Belt.

The Gyeonggi Block has been correlated with the South China Block, whereas the Ryeongnam and Nangrim Blocks have been correlated with the North China Block [2]. According to paleontological data, the Gyeonggi Block has Cambro–Ordovician faunas with South China affinity, while the Ryeongnam and Nangrim Blocks have Cambro–Ordovician faunas with North China affinity [3,4]. It thus appears that the Korean Peninsula has areas of both North and South China

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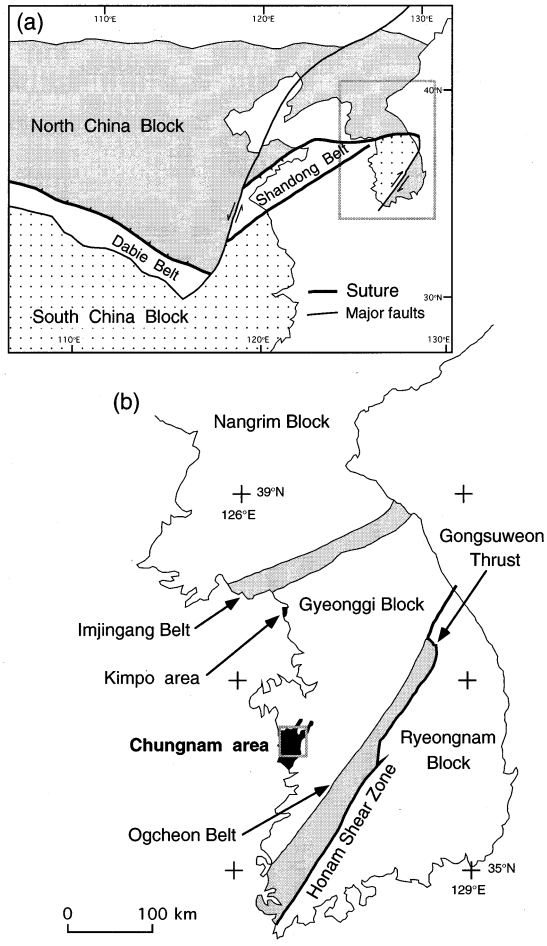


Fig. 1. (a) Tectonic map of the Korean Peninsula and adjacent area. (b) Tectonic units in the Korean Peninsula. The Gyeonggi Block is interpreted to be the eastern part of the South China Block. The Chungnam area (study area) is situated in the southwestern part of the Gyeonggi Block.

affinities: it may be a collage of three continental fragments (Fig. 1a).

Paleomagnetic studies from the Ryeongnam Block, the southeastern sector of the Korean Peninsula, have revealed that the Ryeongnam Block is an eastern extension of the North China Block [5,6]. A paleomagnetic pole for the Ryeongnam Block shows good agreement with that for the North China Block, with the exception of a small clockwise rotation of the Ryeongnam Block with respect to the North China Block [7]. The Ryeongnam Block is now geographically isolated from the main part of the North China Block; however, it is still part of the North China Block.

A few paleomagnetic studies of the Gyeonggi Block, the central part of the Korean Peninsula, have been carried out to investigate its tectonic affinity [8,9]. Paleomagnetic poles from this region lie close to coeval poles from the South China Block [8,9]. The Gyeonggi Block is thus considered to be the eastern part of the South China Block. However, most of the formation-mean poles of those studies (five out of six formation-mean poles) were derived from directions of individual samples because of the small number of samples with a characteristic component. To build up confidence regarding the tectonic affinity of the Gyeonggi Block requires more reliable paleomagnetic results with a sufficient number of samples and positive statistical tests.

The present study aims to obtain a reliable paleomagnetic pole from the Daedong (Taedong) Supergroup and thus to identify the origin of the Gyeonggi Block in early Mesozoic time from a paleomagnetic viewpoint. The paleomagnetic pole from the Gyeonggi Block is compared with those from the North and South China Blocks in order to test the tectonic affinity of the Gyeonggi Block. We focus on the Amisan, Jogyeri and Baegunsa Formations in the Late Triassic to Early Jurassic Daedong Supergroup. These formations are of a strategic age, because early Mesozoic paleomagnetic poles from the North and South China Blocks are separate from each other [10]. The study area is the Chungnam area located in the southwestern part of the Gyeonggi Block (Fig. 1b). In this paper we present a remanence carried by hematite, which is the first report from the Chungnam area.

2. Geological setting and sampling

The Daedong Supergroup is well exposed in the Chungnam and Kimpo areas (Fig. 1b). This supergroup in the Chungnam area is clastic sediments deposited in a freshwater basin, and it unconformably overlies the Precambrian basement rocks of the Gyeonggi Block. The Daedong Supergroup with a total thickness of about

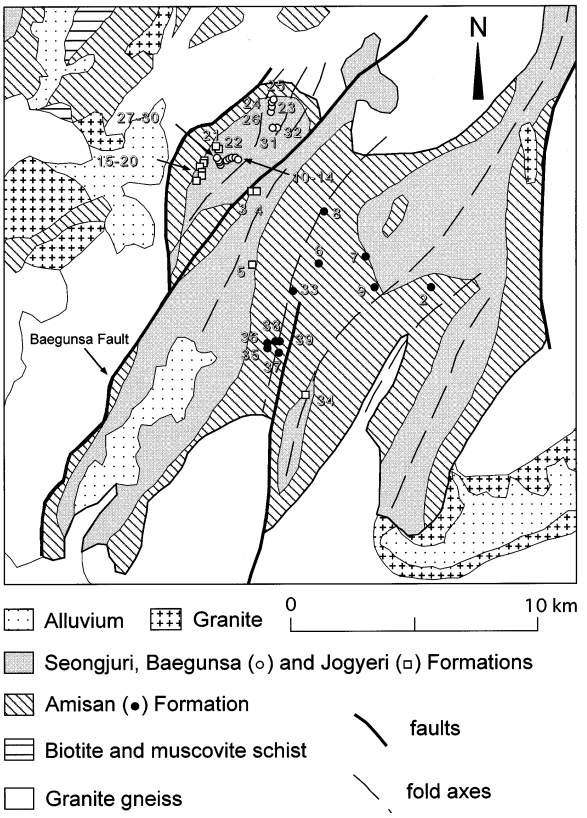


Fig. 2. Geological map of the Daedong Supergroup in the Chungnam area after Reedman and Um [1]. Solid circles, open squares and open circles denote the paleomagnetic sampling sites of the Amisan, Jogyeri and Baegunsa Formations, respectively.

2600 m is divided into five formations: the Hajori, Amisan, Jogyeri, Baegunsa and Seongjuri Formations in ascending order [11,12]. It is made up of shales, sandstones, conglomerates, and coal beds. The Amisan and Baegunsa Formations contain Late Triassic floras of *Baiera* and *Podozamites*. The Amisan Formation also yields a fauna of *Estherites*, which is indicative of the Late Triassic [13]. Although the Daedong Supergroup has an age of Late Triassic–Early Jurassic [1,11,12], the Amisan, Jogyeri and Baegunsa Formations are regarded as of Late Triassic age.

The faults and folds of the Daedong Supergroup were formed during the Daebo orogeny [1]. The Daebo orogeny took place after the deposition of the Daedong Supergroup and ceased with the deposition of the Early Cretaceous Gyeongsang Supergroup on the basis of stratigraphic and structural data, so that the Middle–Late Jurassic age of the faulting and folding is established [14]. In the Chungnam area, the fault and fold axes trend NNE–SSW (Fig. 2). The fault motion must have occurred slightly later than the formation of the folding structure, because the faults cut the folds.

Three hundred and four hand samples of the Daedong Supergroup were collected at 38 sites from the Amisan, Jogyeri and Baegunsa Formations for paleomagnetic measurements (Fig. 2, Table 1). The samples consist of black shales, gray sandstones, and dark gray sandstones. The sampling sites have monoclinal structures with dips ranging from 14° to 78°. Eight hand samples were collected over distances of up to 15 m in each site and oriented with a magnetic compass. The present geomagnetic field declination was calculated using the International Geomagnetic Reference Field [15].

3. Paleomagnetism

In the laboratory, individual samples of 25 mm diameter and 22 mm long were prepared from each hand sample. Natural remanent magnetizations (NRMs) were measured with a 2-G cryogenic magnetometer. All samples were subjected to progressive thermal demagnetization treatment with a Natsuhara TDS-1 thermal demagnetizer.

Results for each sample were plotted on an orthogonal vector diagram [16] to evaluate the demagnetization behaviors. Principal component analysis [17] was used to estimate the directions of the magnetic components (Table 2).

Thermal demagnetization reveals two types of remanent magnetization component. At four sites (TD16, 17, 18 and 19, see Fig. 3a–f), a magnetization component with shallow inclinations and

Table 1
Paleomagnetic sampling details of the Daedong Supergroup

Site	Rock type	*N*	Formation	Locality		Strike	Dip
				Lat. (°N)	Long. (°E)	(°)	(°)
TD2	gray ss.	8	Amisan	36.285	126.750	231	47
TD3	gray ss.	8	Jogyeri	36.323	126.653	41	78
TD4	gray ss.	8	Jogyeri	36.323	126.655	27	72
TD5	dark gray ss.	8	Jogyeri	36.293	126.662	194	71
TD6	gray ss.	8	Amisan	36.295	126.690	248	49
TD7	gray ss.	8	Amisan	36.298	126.716	75	51
TD8	gray ss.	8	Amisan	36.315	126.693	61	31
TD9	gray ss.	8	Amisan	36.285	126.720	234	61
TD10	black shale	8	Baegunsa	36.336	126.645	325	19
TD11	black shale	8	Baegunsa	36.337	126.644	298	31
TD12	dark gray ss.	8	Baegunsa	36.337	126.643	320	26
TD13	black shale	8	Baegunsa	36.336	126.641	301	18
TD14	black shale	8	Baegunsa	36.341	126.636	316	39
TD15	dark gray ss.	8	Jogyeri	36.328	126.625	12	52
TD16	black shale	8	Jogyeri	36.330	126.628	25	73
TD17	black shale	8	Jogyeri	36.331	126.628	26	75
TD18	black shale	8	Jogyeri	36.333	126.627	34	43
TD19	black shale	8	Jogyeri	36.334	126.629	25	48
TD20	black shale	8	Jogyeri	36.335	126.629	17	69
TD21	black shale	8	Jogyeri	36.340	126.637	37	70
TD22	dark gray ss.	8	Jogyeri	36.336	126.637	14	55
TD23	dark gray ss.	8	Baegunsa	36.357	126.665	161	33
TD24	black shale	8	Baegunsa	36.358	126.665	177	52
TD25	black shale	8	Baegunsa	36.359	126.666	174	60
TD26	dark gray ss.	8	Baegunsa	36.355	126.665	159	51
TD27	dark gray ss.	8	Baegunsa	36.334	126.638	359	62
TD28	dark gray ss.	8	Baegunsa	36.335	126.639	344	72
TD29	black shale	8	Baegunsa	36.335	126.640	340	53
TD30	black shale	8	Baegunsa	36.335	126.641	333	43
TD31	black shale	8	Baegunsa	36.348	126.666	185	34
TD32	black shale	8	Baegunsa	36.348	126.668	149	23
TD33	black shale	8	Amisan	36.284	126.676	14	41
TD34	dark gray ss.	8	Jogyeri	36.243	126.679	199	47
TD35	dark gray ss.	8	Amisan	36.260	126.669	211	43
TD36	black shale	8	Amisan	36.261	126.663	268	14
TD37	black shale	8	Amisan	36.263	126.663	230	40
TD38	black shale	8	Amisan	36.264	126.669	219	41
TD39	black shale	8	Amisan	36.264	126.668	220	41

ss = sandstone; *N* = the number of the samples collected; Lat. = latitude; Long. = longitude; Strike = right-hand strike.

Table 2
Paleomagnetic results of the Daedong Supergroup

Site	N (n/r)	In situ		Tilt-corrected		k	α_{95}
		D (°)	I (°)	D (°)	I (°)		(°)
Component A							
TD16	7 (6/1)	16.1	58.4	83.9	22.2	241.3	3.9
TD17	8 (8/0)	6.9	55.9	80.6	22.9	37.7	9.1
TD18	6 (0/6)	238.2	−46.3	261.9	−19.5	271.8	4.1
TD19	7 (3/4)	223.3	−60.0	262.3	−27.6	32.2	10.8
Mean	4	32.6	57.0			29.2	17.3
				82.2	23.1	508.3	4.1
Component B							
TD2	5 (5/0)	10.3	61.4	343.9	21.8	609.5	3.1
TD13	3 (3/0)	0.8	57.9	10.0	41.5	172.5	9.4
TD14	6 (6/0)	359.8	55.4	19.4	23.0	926.4	2.2
TD24	3 (3/0)	349.1	57.5	303.6	27.6	743.1	4.5
TD26	5 (5/0)	352.8	59.9	288.0	39.7	252.0	4.8
Mean	5	358.4	58.6			288.0	4.5
				338.2	36.0	5.7	35.2

N = number of the samples in calculation for the means; (n/r) = the number of samples with normal/reversed polarity; D and I = declination and inclination, respectively; k = the precision parameter; α_{95} = radius of the cone of the 95% confidence.
Mean direction is calculated at the Gyeonggi Block reference point (36.3°N, 126.7°E).

dual polarities after tilt correction is isolated in the form of higher temperature trajectories (hereafter referred to as component A). At five sites (TD2, 13, 14, 24 and 26, see Fig. 3g), a magnetization component with directions conforming to the present Earth's field before tilt correction is observed in the form of univectorial behavior (hereafter referred to as component B). The remaining 29 sites exhibit either erratic directional behavior with no stable end points or an unstable magnetization component removed by 150°C (Fig. 3h).

3.1. Component A

Component A decays straightforwardly to the origin. This component is generally resolved by demagnetization steps above 580°C, where more than 50% of the initial NRM persists. Above this temperature, a large decay of the remanent magnetization is observed. The component is unblocked between 635 and 660°C, indicating that it resides in hematite.

The component A in site TD17 is of normal polarity, while this component in site TD18 is of reversed polarity. The component A in sites TD16 and 19 contains reversals: site TD16 has one reversed direction out of seven, and site TD19 has four reversed directions out of seven. The reversal test of McFadden and McElhinny [18] was performed between the samples of normal and reversed polarities: 17 normal directions and 11 reversed directions. A positive reversal test (classification B) at the 95% confidence level is yielded after tilt correction, whereas an inconclusive reversal test is yielded before tilt correction. The angular distance between the mean directions of the two polarities after tilt correction is 6.4°, and the critical angle is 7.1°.

Component A was obtained from the four sites of the Jogyeri Formation in the Daedong Supergroup (sites TD16, 17, 18 and 19). These sites have roughly consistent bedding strikes of N25–34°E with dips ranging from 43 to 75°SE (Table 1), and constitute a stratigraphic interval of about 200 m. Three magnetostratigraphic polarity zones are identified within this interval: one normal and two reversed polarity zones.

For each site, directions of component A are averaged in a hemisphere best suited to calculate a site-mean direction (Table 2). The four site-mean directions cluster well at each site: the precision parameter of the four directions ranges from 32.2 to 271.8. The site-mean directions

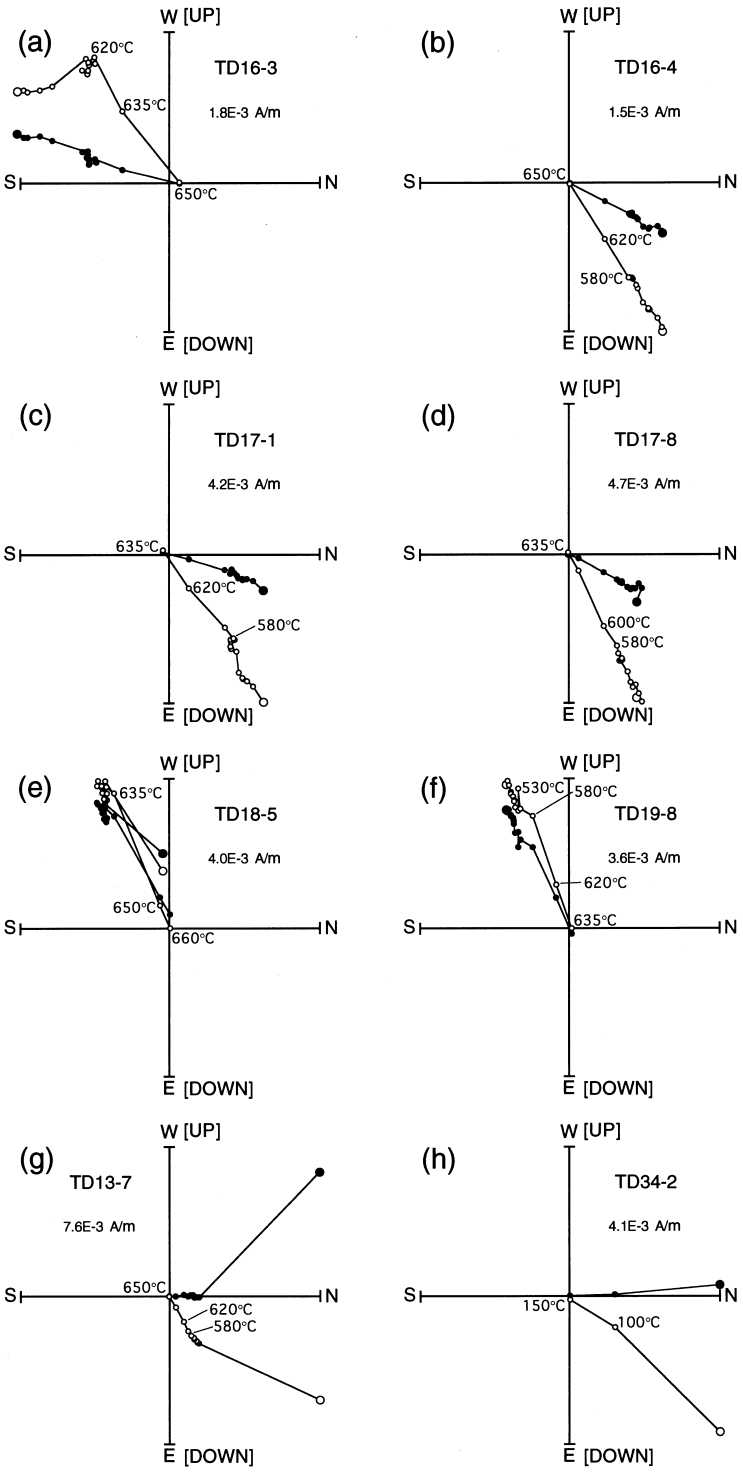


Fig. 3. Orthogonal plots of thermal demagnetization of the Daedong Supergroup (before tilt correction). Solid and open symbols represent projection on horizontal and vertical planes, respectively.

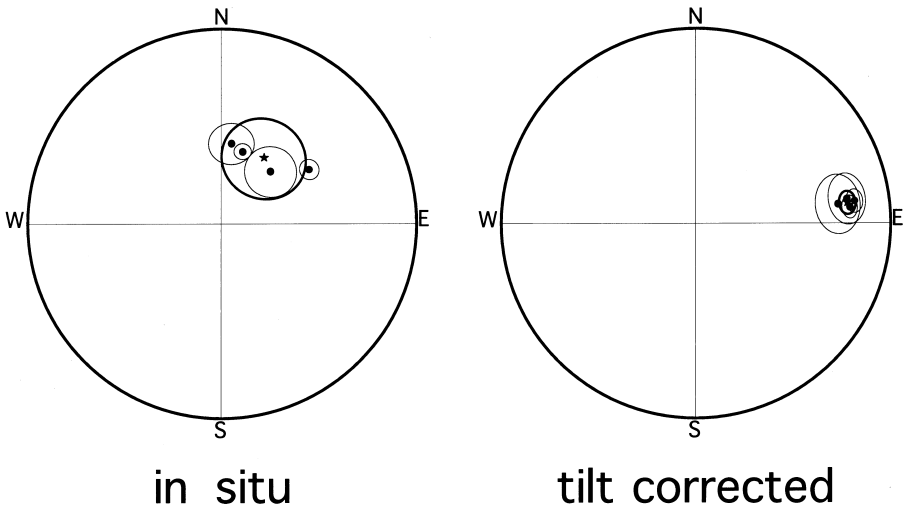


Fig. 4. Equal area projections of the directions of component A (with associated 95% confidence limits) from four sites of the Jogyeri Formation in the Daedong Supergroup. All the directions are plotted in the lower hemisphere in order to show the clustering of the directions. Star and bold circle represent the mean direction and associated 95% confidence limit, respectively. The fold test is positive at the 95% confidence level [19].

show a fairly tight clustering after tilt correction (Fig. 4). The precision parameter increases from 29.2 to 508.3. The fold test is positive at the 95% confidence level [19], indicating that component A is of pre-folding origin. The component A with both normal and reversed polarities is thus interpreted to be of primary origin. The folding age of the Middle–Late Jurassic [14] supports that this component is essentially of primary origin. The mean direction of component A is $D = 82.2°$, $I = 23.1°$ with $\alpha_{95} = 4.1°$.

3.2. Component B

Component B in site TD2 is unblocked at 350°C, with a large decay at temperatures above 300°C, suggesting that the remanent magnetization is carried by pyrrhotite. The component B

Table 3
Late Triassic paleomagnetic poles for the Gyeonggi, South China and North China Blocks

Locality	N	Pole position		A_{95}	Reference
(Formation)		Lat. (°N)	Long. (°E)	(*dp*/*dm*) (°)	
Gyeonggi Block					
(Seongjuri)	6*	34.4	225.9	(5.6/10.5)	[9]
(Baegunsa)	6*	13.7	208.8	(3.1/5.7)	[9]
(Jogyeri)	8*	42.7	236.4	(4.3/8.1)	[9]
(Amisan)	9*	28.7	228.8	(4.1/7.9)	[9]
(Munjusan)	5	47.8	227.9	(4.2/7.2)	[8]
(Tongjin)	14*	39.6	207.5	(7.0/10.6)	[8]
(Jogyeri)	4	13.3	211.4	2.7	This study
Mean	7	31.9	220.2	12.7	
South China Block	23*	44.8	223.6	9.3	[20]
North China Block	11	62.3	7.7	3.8	[21]

N = the number of the sites or *samples; Lat. = latitude; Long. = longitude; A_{95} = radius of the circle of 95% confidence.

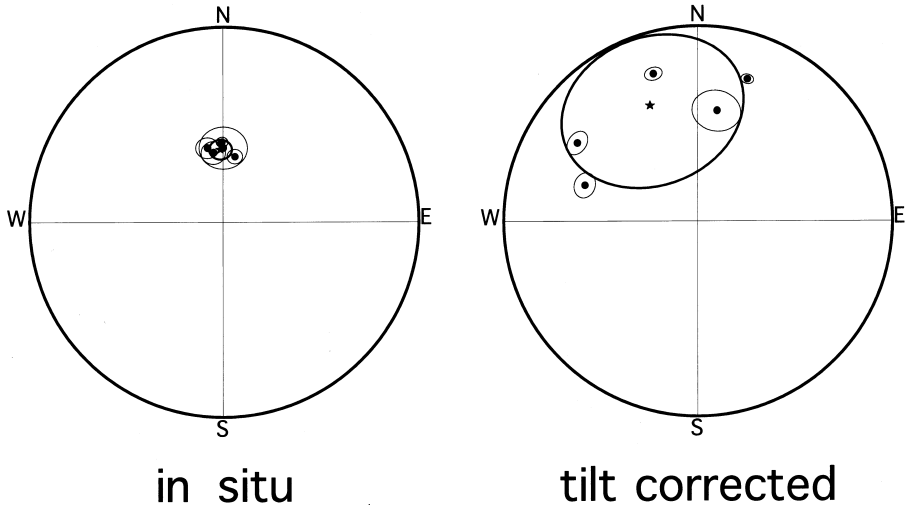


Fig. 5. Equal area projections of the directions of component B (with associated 95% confidence limits) from five sites of the Daedong Supergroup. Star and bold circle represent the mean direction and associated 95% confidence limit, respectively. The fold test is negative at the 95% confidence level [19].

in the other four sites with unblocking temperatures of 620–650°C shows a steady decay until 580°C and a large decay above this temperature. Little change is observed in the direction of the remanent magnetization after demagnetization at temperatures higher than the magnetite Curie temperature of 580°C. The remanent magnetization resides in magnetite and hematite with almost the same direction.

Component B is uniformly of normal polarity. The five site-mean directions of this component have a tightly clustered mean of $D = 358.4°$, $I = 58.6°$ with $\alpha_{95} = 4.5°$ before tilt correction (Fig. 5). The tight clustering disappears after tilt correction: the precision parameter decreases from 288.0 to 4.5. These directions yield a negative fold test at the 95% confidence level [19], indicating that component B is of post-folding origin. Component B is ascribed to a recent remagnetization.

4. Discussion

The previous paleomagnetic data of the Daedong Supergroup from the Chungnam area were reported by Kim and Kim [9], and those from the Kimpo area by Kim et al. [8] (Fig. 1b, Table 3). The Chungnam area provided a characteristic component from four formations [9]. Three formations out of the four (Seongjuri, Baegunsa and Jogyeri Formations) possessed both normal and reversed directions and yielded a positive fold test. The other formation in the Chungnam area (Amisan Formation) showed normal directions and the fold test was not quite significant, but the paleomagnetic directions of this formation are similar to those of the three formations. Two formations in the Kimpo area gave a characteristic component with both normal and reversed polarities [8]. One formation (Munjusan Formation) showed a positive fold test. Although another formation (Tongjin Formation) yielded inconclusive fold test, no significant difference in paleomagnetic direction exists between the two formations.

To obtain a mean paleomagnetic pole of the Daedong Supergroup which is representative of the Gyeonggi Block, we averaged all seven available poles of the Daedong Supergroup [8,9] (Table 3). These poles from the Gyeonggi Block are collected from two localities in the undisputed part of the block: the Chungnam and Kimpo areas (Fig. 1b). Local tectonic rotation due to fault motion in the Chungnam area is considered unlikely, because the previous results of this area [9] demonstrated consistency of paleomagnetic di-

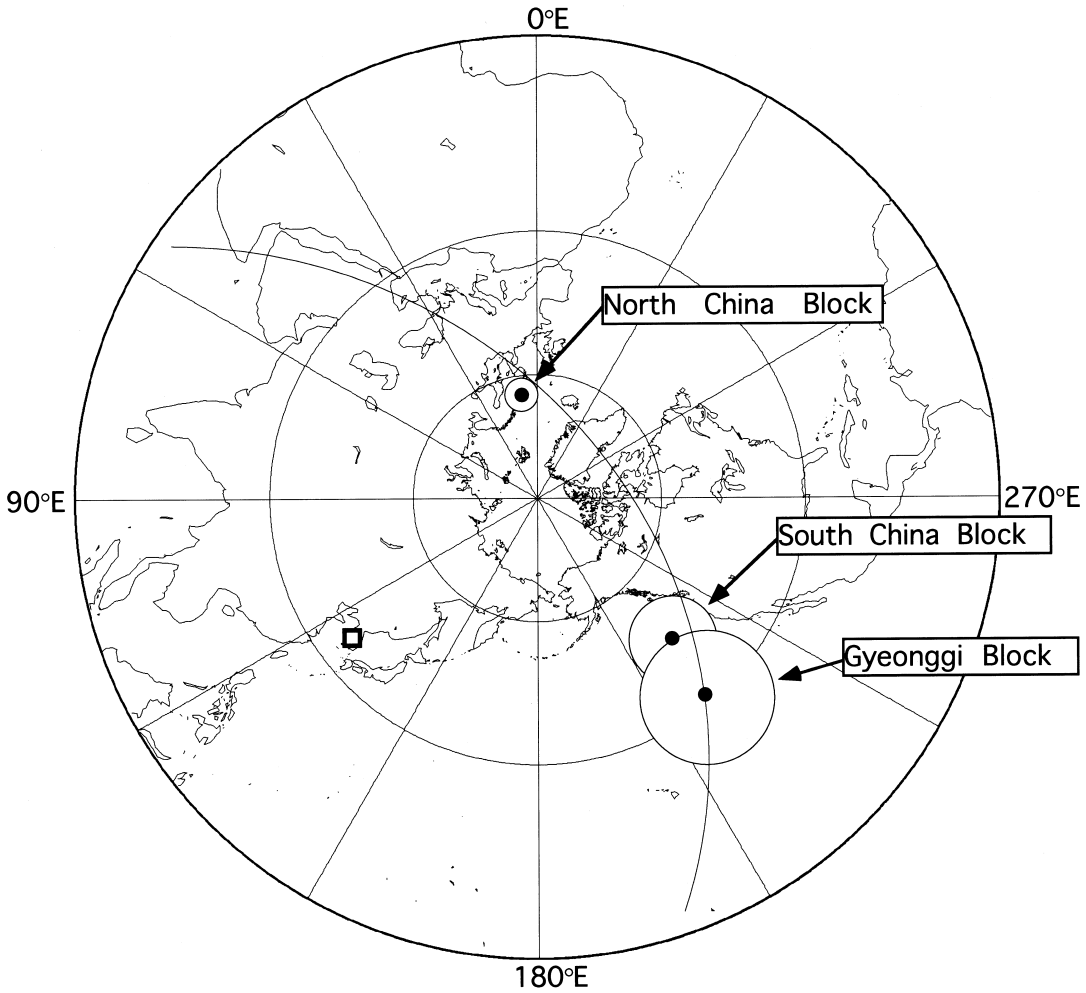


Fig. 6. Late Triassic paleomagnetic poles (with associated 95% confidence limits) for the Gyeonggi Block, South China Block and North China Block. Open square denotes the reference point of the Gyeonggi Block. Small circle is centered on the reference point. Data from Table 3.

rections between areas east and west of the Baegunsa Fault (Fig. 2). Local rotation in the Kimpo area is also unlikely, because there are no large faults that cut the Daedong Supergroup in that area [1]. Because (1) these poles pass reliability tests, (2) the poles are calculated from both normal and reversed directions, and (3) the poles cover a distance of about 200 km in the Gyeonggi Block, the mean pole (31.9°N, 220.2°E with $A_{95} = 12.7°$) is regarded as representative of this block.

The paleomagnetic pole for the Gyeonggi Block is compared with the Late Triassic pole for the South China Block at 44.8°N, 223.6°E with $A_{95} = 9.3°$ [20]. The similarity of pole positions is apparent: the angular difference between the poles is $13.1 \pm 15.7°$ (Fig. 6). In contrast to the case for the South China Block, the Late Triassic pole for the North China Block (62.3°N, 7.7°E with $A_{95} = 3.8°$) [21] is located far from that for the Gyeonggi Block. The angular distance between the poles is $82.2 \pm 13.2°$. This clearly shows that the Gyeonggi Block has been part of the South China Block. Paleontological data from the Gyeonggi Block corroborate this correlation: the Daedong Supergroup in the study area yields

the Late Triassic *Dictyophyllum-Clathropteris* floras that are found in a wide area of the South China Block [22]. Paleozoic paleontological data from the Gyeonggi Block also suggest the South China affinity of this block [3,4].

Although the difference in pole positions between the Gyeonggi and South China Blocks is not quite significant, the pole for the Gyeonggi Block is plotted to a lower latitudinal position than that for the South China Block. The difference of pole positions is described as a possible clockwise rotation of $13.1 \pm 20.4°$ (with a latitudinal difference of $5.1 \pm 15.7°$) of the Gyeonggi Block with respect to the South China Block, calculated at the reference point of the Gyeonggi Block (36.3°N, 126.7°E). This clockwise sense of rotation for the Gyeonggi Block can be explained by the post-Cretaceous clockwise rotation (10–36°) of the area east of the Tan-Lu Fault with respect to the main part of the integrated China blocks (North and South China Blocks) [23,24].

The Korean Peninsula is tectonically composed of elements of both North and South China Blocks, because (1) the paleomagnetic pole for the Gyeonggi Block indicates that the Gyeonggi Block has behaved as an integral part of the South China Block, (2) the previously reported paleomagnetic pole for the Ryeongnam Block, the southeastern part of the Korean Peninsula, demonstrates that the Ryeongnam Block has been part of the North China Block [7], (3) no significant paleolatitude difference is observed between the Ryeongnam and North China Blocks through Paleozoic times at the reference point within the Ryeongnam Block [6], and (4) the Nangrim Block, the northern part of the peninsula, is a part of the North China Block because of the geographically and geologically direct connection of this area with the main part of the North China Block. This paleomagnetic correlation for each block is fairly consistent with that from Paleozoic paleontological data [3,4].

The Korean Peninsula has been involved in the collision event between the North and South China Blocks as their easternmost parts. Paleomagnetic researches revealed that the collision between the two blocks occurred initially in their eastern parts and progressed westward [10,25]. The mode of the collision is explained by the scissor-like process that includes counterclockwise rotation of the North China Block and clockwise rotation of the South China Block with a rotation pole in their eastern part. Since the easternmost part of the North and South China Blocks is in the Korean Peninsula, the collision between the North and South China Blocks commenced initially at the Korean Peninsula.

Li [26] suggested that the upper crust of the eastern part of the South China Block was detached from the lower crust of this area, and thrust over the North China Block. The Gyeonggi Block as the eastern part of the South China Block could have been obducted onto the eastern part of the North China Block (site of the present Korean Peninsula) during this process. The Gyeonggi Block was conveyed to the present position on the possible detachment with a likely depth of 20 km, estimated from seismic results [26]. According to structural analysis of Reedman and Um [1], the two Paleozoic bioprovinces of the North and South China Blocks have a thrust contact within the Korean Peninsula, where the sequence of the South China bioprovince structurally overlies that of the North China bioprovince along the Gongsuweon Thrust (also referred to as the Mungyeong-Jeongseon Tectonic Line) (Fig. 1b). The thrust is probably the product of the obduction. This model is different from the 'lithospheric' indentation model of Yin and Nie [27].

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